

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03/08/2026	Duration:	60 minutes

Code of Conduct

I D.G.RAGHAVI (192521173) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: D.G.RAGHAVI

Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. Analyze System Requirements

- Study the given problem statement and identify the system objectives.
- Analyze the functional and non-functional requirements that influence the software architecture.
- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.
- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design	Produces a clear, complete, and well-structured architecture diagram with all	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and	Diagram is incomplete, unclear, or

	major components, interfaces, and interactions accurately represented.		lacks sufficient detail.	technically incorrect.
Component Interaction and Quality Attribute Analysis	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	Mark
System Requirement Analysis	/5
Selection of Software Architecture	/5
Architecture Diagram and Component Design	/5
Component Interaction and Quality Attribute Analysis	/5
Architectural Justification and Technical Presentation	/5
TOTAL	/5

1.1 System Objectives

The AI-Based Smart Construction Project Monitoring and Risk Assessment Platform is designed to continuously monitor construction activities using Artificial Intelligence (AI), Internet of Things (IoT), drones, cameras, and Building Information Modeling (BIM). The system analyzes real-time site data to detect safety violations, predict project delays, optimize resource utilization, assess construction risks, and generate project performance dashboards, enabling informed decision-making and improving project execution.

Objectives

- Monitor construction progress continuously.
- Detect safety violations using AI.
- Predict project delays.
- Optimize resource allocation.
- Track equipment utilization.
- Generate project performance dashboards.
- Assess construction risks.
- Improve project management efficiency.

1.2 Functional Requirements

User Management

- User registration and login.
- Role-based access (Project Manager, Site Engineer, Safety Officer, Admin).
- User profile management.

Construction Site Monitoring

- Capture images and videos from drones and CCTV cameras.
- Collect real-time sensor data (temperature, vibration, equipment status).
- Monitor worker attendance and activity.

AI Risk Assessment

- Detect safety violations (missing helmets, safety vest violations, unsafe zones).
- Identify potential hazards using computer vision.

- Predict project delays using AI models.
- Assess construction risks based on historical and real-time data.

Resource Management

- Monitor equipment utilization.
- Track labor productivity.
- Allocate resources efficiently.

Progress Tracking

- Compare actual progress with planned schedules.
- Monitor task completion percentages.
- Generate daily, weekly, and monthly reports.

Dashboard & Reporting

- Display project progress.
- Show safety alerts.
- Generate risk assessment reports.
- Visualize KPIs and analytics.

Notification System

- Safety violation alerts.
- Equipment maintenance reminders.
- Project delay notifications.
- Emergency alerts.

Administration

- Manage users.
- Configure AI models.
- Manage project and sensor information.

1.3 Non-Functional Requirements

Performance

- Real-time monitoring with response time below 3 seconds.
- Efficient AI image processing.

Security

- Secure login with JWT/OAuth authentication.
- Encrypted communication using HTTPS.
- Role-based access control.
- Secure cloud storage.

Reliability

- Continuous monitoring (24×7).
- Automatic backup and recovery.
- Fault-tolerant architecture.

Scalability

- Support multiple construction projects simultaneously.
- Easily add new drones, cameras, and IoT devices.

Maintainability

- Modular software architecture.
- Easy AI model updates.
- Independent service maintenance.

Availability

- Cloud deployment with high availability.
- Load balancing and failover support.

Usability

- User-friendly dashboard.
- Mobile and web accessibility.

1.4 Quality Attributes

Quality Attribute	Description
Scalability	Supports multiple projects, sensors, and users through independent service scaling.
Security	Protects project and monitoring data using authentication, encryption, and secure APIs.

Quality Attribute	Description
Performance	Processes drone and sensor data in real time for quick decision-making.
Reliability	Ensures uninterrupted monitoring through fault isolation and backups.
Maintainability	Modular architecture enables easy updates and maintenance.
Availability	Cloud infrastructure provides high uptime and disaster recovery.

2. Software Architecture Selection

Selected Architecture

Hybrid Architecture (Microservices + Event-Driven + Layered Architecture)

Why Hybrid Architecture?

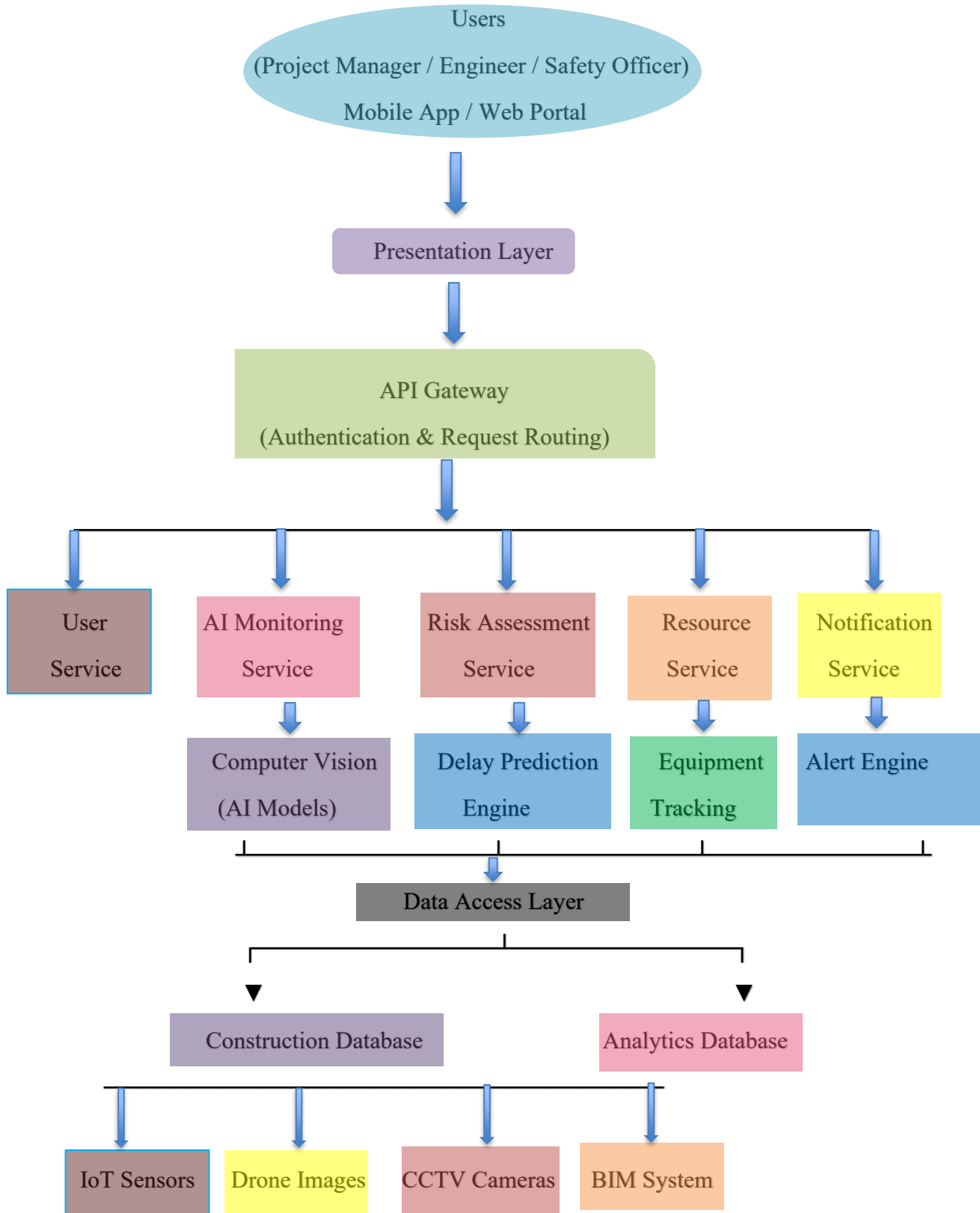
The proposed system combines **Microservices**, **Event-Driven**, and **Layered Architecture**.

- **Microservices** separate major functions such as AI Monitoring, Risk Assessment, Resource Management, Notifications, and Reporting into independent services.
- **Event-Driven Architecture** enables real-time processing of events generated by drones, IoT sensors, and cameras.
- **Layered Architecture** organizes each service into Presentation, Business Logic, and Data Access layers for maintainability.

Justification

- Supports continuous real-time monitoring.
- Independent AI services improve scalability.
- Efficient event processing from IoT devices.
- Simplifies maintenance and testing.
- Secure communication using API Gateway.
- Easy integration with BIM, drones, cameras, and IoT platforms.
- Supports future smart construction technologies.

3. Software Architecture Diagram



4. Components and Their Interactions

Components and Responsibilities

Component	Responsibility
Presentation Layer	Provides web and mobile dashboards for project monitoring.
API Gateway	Authenticates users, routes requests, and secures communication.
User Service	Manages users, authentication, and project roles.
AI Monitoring Service	Analyzes images and videos to monitor project progress and detect safety violations.
Risk Assessment Service	Predicts project delays and evaluates construction risks using AI.
Resource Management Service	Tracks labor, equipment, and material utilization.
Notification Service	Sends safety alerts, maintenance reminders, and project notifications.
Construction Database	Stores project details, worker information, equipment data, and monitoring records.
Analytics Database	Stores historical reports, KPIs, AI predictions, and dashboards.
External Systems	Integrates drones, IoT sensors, CCTV cameras, and BIM software.

Component Interaction Workflow

1. Users access the platform through the web or mobile application.
2. The API Gateway authenticates users and forwards requests.
3. Drones, CCTV cameras, IoT sensors, and BIM systems continuously send construction data.
4. The AI Monitoring Service processes images and videos using computer vision.
5. The Risk Assessment Service predicts delays and identifies project risks.
6. The Resource Management Service tracks equipment and labor utilization.
7. All processed data is stored in the Construction and Analytics Databases.
8. The Notification Service sends real-time safety alerts and project updates.

9. Dashboards display project progress, safety compliance, resource usage, and risk analysis.

5. Quality Attribute Evaluation

Quality Attribute	Implementation
Scalability	Independent microservices allow AI, monitoring, and reporting services to scale individually for multiple projects.
Security	JWT authentication, HTTPS, encrypted databases, and role-based access control secure project information.
Performance	Event-driven processing and AI image analysis provide real-time monitoring and quick responses.
Reliability	Fault isolation, cloud backups, and redundant services ensure continuous monitoring.
Maintainability	Layered architecture and modular microservices simplify updates, testing, and maintenance.
Availability	Cloud deployment with load balancing and failover ensures uninterrupted access.

6. Architectural Decisions

Rationale Behind Choosing Hybrid Architecture

The **Hybrid Architecture (Microservices + Event-Driven + Layered)** is selected because construction monitoring systems require continuous real-time data processing, AI-based analysis, and independent service scalability.

Reasons

- Supports AI-based safety monitoring and risk prediction.
- Processes real-time events from drones, IoT sensors, and cameras.
- Enables independent deployment and scaling of monitoring services.
- Simplifies maintenance through modular design.
- Integrates seamlessly with BIM, IoT platforms, and cloud services.
- Provides fault isolation to improve reliability.

Trade-offs Considered

Advantages	Trade-offs
Real-time monitoring	Higher deployment complexity
Independent service scalability	Increased inter-service communication
Fault isolation	Requires event brokers and API management
Easy future expansion	Higher infrastructure and operational cost
Better maintainability	More complex monitoring and orchestration

Future Enhancements and Long-Term Maintainability

The architecture supports future expansion by allowing new services to be added without affecting existing components.

Possible future enhancements include:

- Digital Twin integration for virtual construction monitoring.
- Autonomous drone inspections.
- Predictive maintenance for construction equipment.
- Augmented Reality (AR) site visualization.
- Integration with ERP and project management systems.
- Advanced AI models for quality inspection and cost prediction.
- Multi-project analytics and benchmarking dashboards.

Because the system is built using modular microservices, event-driven communication, and layered design, it remains scalable, maintainable, and adaptable to future smart construction technologies.

Conclusion

The Hybrid Architecture (Microservices + Event-Driven + Layered Architecture) is the most suitable architectural style for the AI-Based Smart Construction Project Monitoring and Risk Assessment Platform. It satisfies both the functional and non-functional requirements by providing a scalable, secure, reliable, high-performance, and maintainable solution. The architecture supports real-time monitoring, AI-driven safety analysis, IoT and drone integration, cloud-based data management, and predictive risk assessment while offering a robust foundation for future enhancements and long-term project success.

